

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	10-08-2026
Team ID	23331A4450
Project Name	Automated Network Request Management System using ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

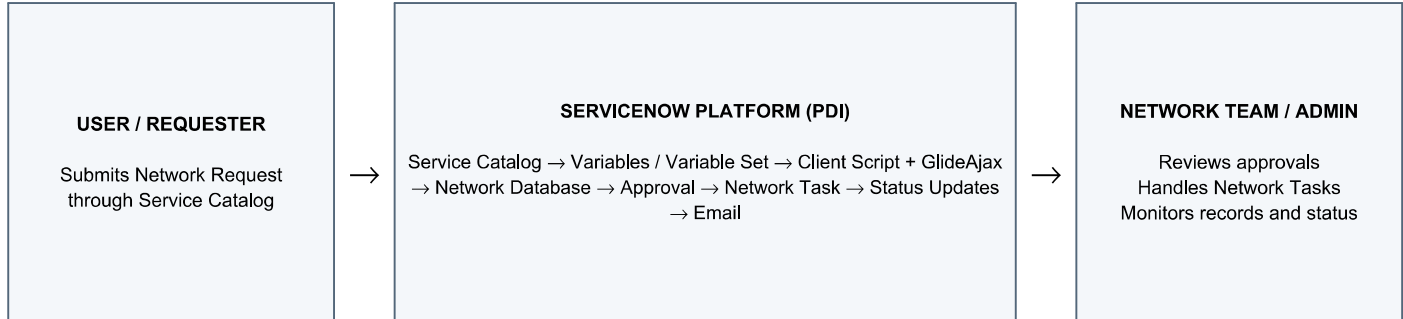
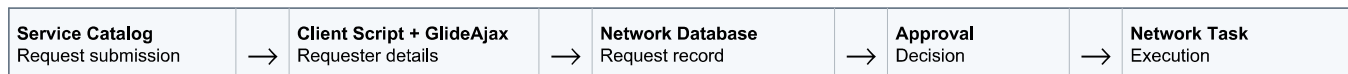


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface used by employees to submit network service requests.	ServiceNow Service Catalog, Service Portal
2	Request Data	Collects requester, device, address, document and request details.	Catalog Variables, Variable Set
3	Client-side Automation	Automatically populates requester information and controls form behavior.	Catalog Client Script, UI Policy
4	Server-side Data Retrieval	Retrieves user details securely from the server.	Script Include, GlideAjax
5	Application Database	Stores submitted network request information and workflow status.	Custom Network Database Table
6	Task Management	Stores work assigned to the network team after approval.	Custom Network Task Table
7	Workflow Automation	Automates record creation, approvals, updates and notifications.	Flow Designer
8	Approval Management	Handles Network Database and Network Task approval decisions.	ServiceNow Approvals / Flow Designer
9	Notifications	Sends confirmation, approval/completion and rejection emails.	Send Email / Notification Engine
10	Security	Controls access to application data and operations.	ACLs, Roles
11	Source Control	Maintains application configuration in version control.	ServiceNow Source Control, GitHub

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Client-side Customization	Improves form usability by populating user details and applying conditional field behavior.	JavaScript, Catalog Client Scripts, Catalog UI Policies
2	Server-side Processing	Separates server-side user-data retrieval from client-side form logic.	Script Includes, GlideAjax
3	Security Implementation	Provides controlled access to application data and functions.	ServiceNow ACLs, User Roles
4	Workflow Automation	Coordinates approvals, task creation, record updates and notifications.	Flow Designer, Approvals
5	Traceability	Maintains separate request and task records with approval and work status.	Network Database, Network Task tables
6	Maintainability	Uses modular ServiceNow configuration components that can be updated independently.	Scoped Application, Source Control
7	Availability	Runs on the ServiceNow cloud platform and is accessible through the configured application interface.	ServiceNow Cloud Platform
8	Notifications	Communicates important workflow outcomes to the requester.	Email / Notification Engine

Technology Flow:**References:**

ServiceNow Service Catalog documentation – <https://www.servicenow.com/docs/> ServiceNow Flow Designer documentation – <https://www.servicenow.com/docs/> ServiceNow Developer documentation – <https://developer.servicenow.com>